

Supplementary Statistical Analysis

This anonymous supplement accompanies 'ASM-Net: Anisotropic Small-hemorrhage Modeling Network for Thick-Slice Intracranial Hemorrhage Segmentation'.

Statistical testing was conducted on the primary cohort only. External supporting benchmarks were not used for formal statistical testing because of their smaller sample sizes and heterogeneous annotation protocols.

Paired Wilcoxon signed-rank tests were used for matched case-level comparisons. Bootstrap 95% confidence intervals used 10,000 resamples with random seed 2026. For Dice, positive differences indicate higher Dice for ACE-Net; for HD95 and ASD, positive differences indicate lower boundary error for ACE-Net.

The table reports improvement magnitudes, bootstrap 95% confidence intervals, and Holm-Bonferroni corrected p-values. Absolute ACE-Net Dice values are intentionally omitted from this compact supplement table.

Paired Statistical Analysis

Positive values favor ACE-Net. Dice and Small Dice are reported as percentage-point improvements; HD95 and ASD are reported as millimeter reductions.

Compared Method	Metric	n	Improvement	95% CI	Holm p
SegMamba	Dice	191	1.67 pp	[-0.05, 3.52] pp	0.504
SegMamba	HD95	191	55.61 mm	[22.62, 91.30] mm	0.019
SegMamba	ASD	191	6.53 mm	[-3.60, 15.29] mm	9.27e-08
SegMamba	Small Dice	38	5.71 pp	[-1.06, 13.19] pp	0.504
MedNeXt	Dice	191	0.48 pp	[-0.80, 1.85] pp	0.359
MedNeXt	HD95	191	20.90 mm	[3.13, 40.85] mm	0.504
MedNeXt	ASD	191	3.95 mm	[-0.15, 8.73] mm	0.291
MedNeXt	Small Dice	38	5.17 pp	[-0.22, 10.96] pp	0.504
UKAN	Dice	191	1.41 pp	[0.31, 2.64] pp	6.82e-06
UKAN	HD95	191	12.34 mm	[-3.08, 29.68] mm	0.431
UKAN	ASD	191	-0.45 mm	[-6.89, 5.03] mm	0.359
UKAN	Small Dice	38	2.18 pp	[-2.38, 6.98] pp	0.504
DynUNet	Dice	191	1.19 pp	[-0.20, 2.66] pp	0.504
DynUNet	HD95	191	73.13 mm	[45.05, 103.84] mm	3.45e-04
DynUNet	ASD	191	10.03 mm	[0.65, 18.89] mm	1.61e-14
DynUNet	Small Dice	38	7.23 pp	[1.07, 13.58] pp	0.091
3D U-Net	Dice	191	3.70 pp	[2.08, 5.48] pp	0.003
3D U-Net	HD95	191	112.90 mm	[77.48, 150.96] mm	1.24e-08
3D U-Net	ASD	191	19.53 mm	[8.92, 29.64] mm	1.03e-16
3D U-Net	Small Dice	38	12.09 pp	[6.13, 18.55] pp	0.018

Reproducibility and Anonymization Notes

The statistical analysis can be reproduced with
`supplementary/statistical_analysis/run_statistical_analysis.py`.

All case labels in the supplementary directory have been anonymized. The supplement does not include author names, hospital names, original case identifiers, DICOM metadata, or local filesystem paths.